

TA212 LAB

PROJECT REPORT, 1ST SEMESTER 2025-26

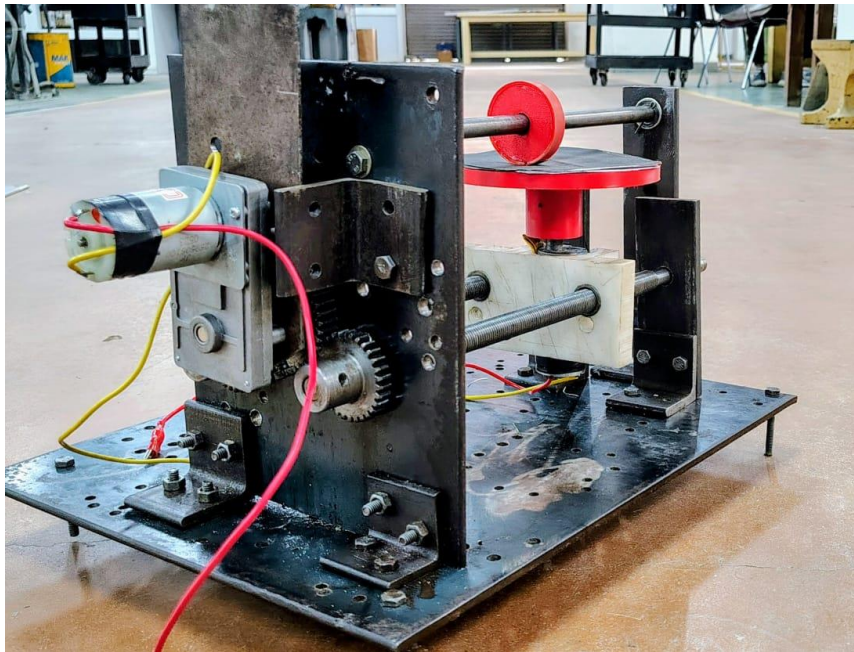
MECHANICAL INTEGRATOR

Group No. – 9

Lab Day – TUESDAY

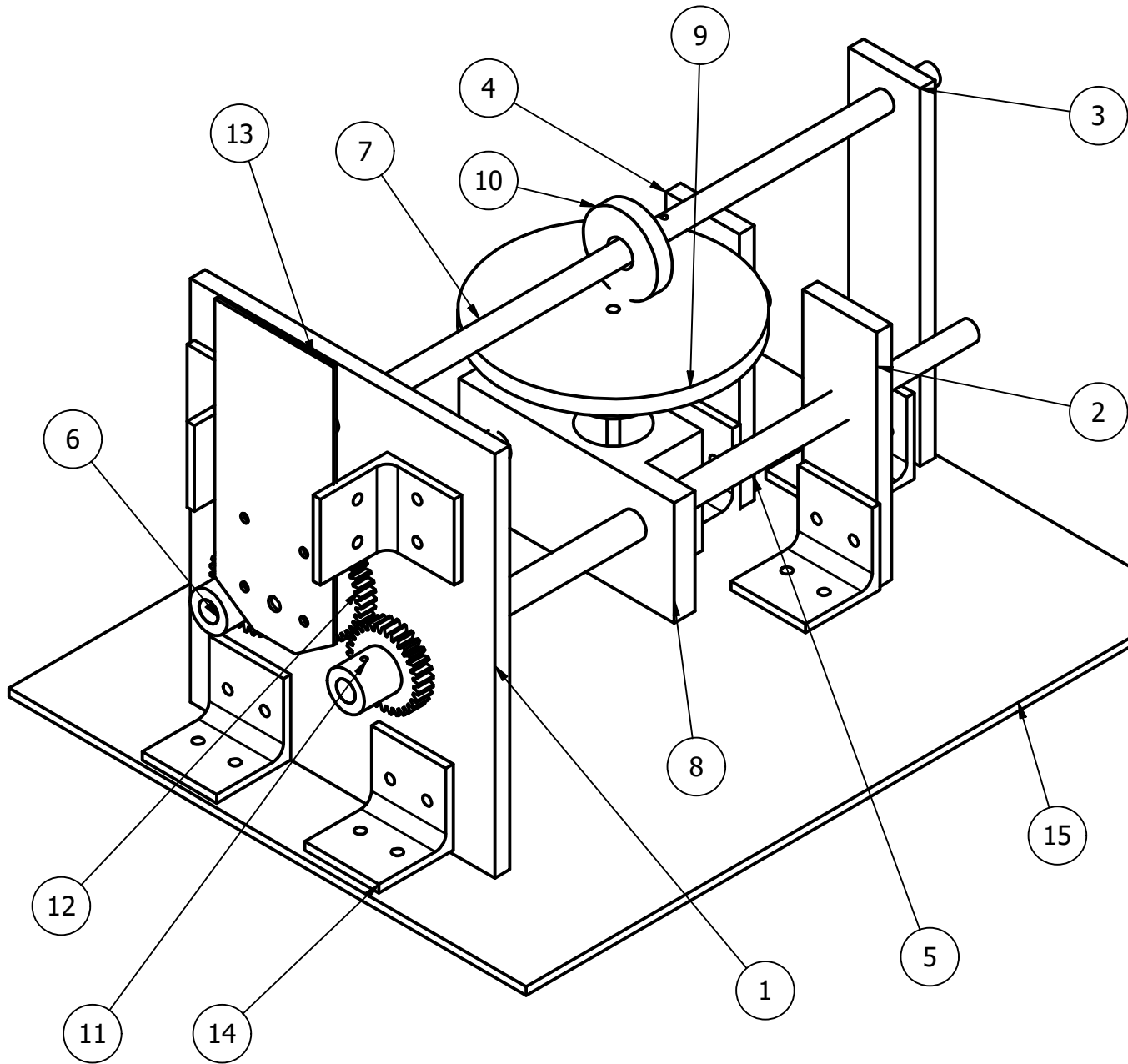
Course Instructor – Dr. Sarvesh Mishra

Lab Guide – Arun Kumar Dubey



Group Members

Sl. No.	Name	Roll No.
1.	Adish Sudhir Kapate	240046
2.	B Mukund Advait	240253
3.	Daksh M Jain	240319
4.	Pratik Beniwal	240783
5.	Priyanshu Mehta	240808
6.	Rohith Rajeev Nambiar	240882
7.	S Kiran	240895
8.	Vegesna Ram Charan Varma	241156



MECHANICAL INTEGRATOR

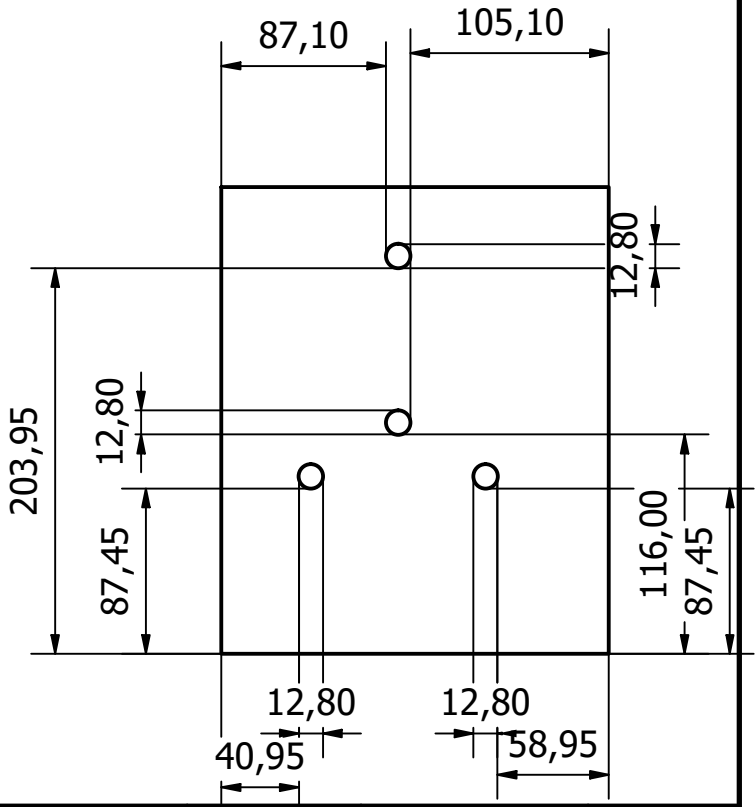
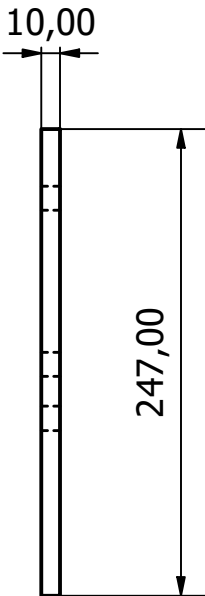
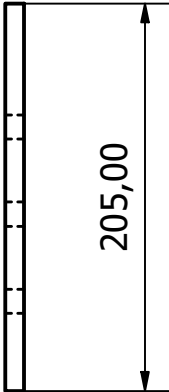
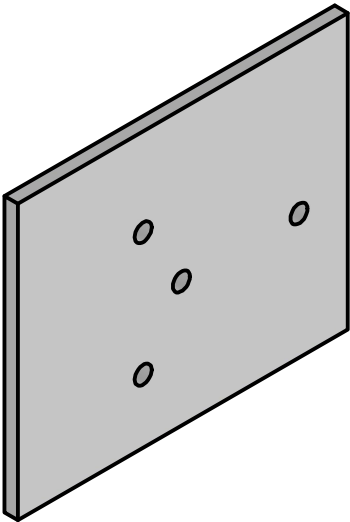
DRAWN Mukund Advaith	11-11-2025	TITLE		
CHECKED				
QA				
MFG				
APPROVED		SIZE A4	DWG NO	REV
		SCALE 1 / 3	SHEET 1 OF 1	

INDEX / PART LIST

Sl. No.	Part No.	Part Name	Qty	Dimensions in mm	Material	Pg. No.
1	-	Cover Page	-	-	-	1
2	-	Complete Assembly	-	-	-	2
3	-	Index / Part List	-	-	-	3
4	1	Big Wall	1		MS	4
5	2	Tall Wall 1	1		MS	5
6	3	Tall Wall 2	1		MS	6
7	4	Tall Wall 3	1		MS	7
8	5	Drive Shaft 1	1		MS	8
9	6	Drive Shaft 2	1		MS	9
10	7	Stationary Rod	1		MS	10
11	8	Carriage	1		Derlin	11
12	9	Big Disk	1		PLA	12
13	10	Small Disk	1		PLA	13
14	11	Spur Gear (n=30)	2		MS	14
15	12	Spur Gear (n=40)	1		MS	15
16	13	Motor Holding Plate	1		MS	16
17	14	L - Clamp	7		MS	17
18	16	Base	1		MS	18
19	-	Project Overview & Motivation	-	-	-	19
20	-	Acknowledgement	-	-	-	20
21	-	Motor Torque Calculations	-	-	-	21
22	-	Log Sheets	-	-	-	22-29
23	-	Cost Estimation	-	-	-	30

Scale-1:2
Dimension-247x205x10
Process-Conventional(Drilling,
milling,turning,lathing)
Material-M.S Sheet(10mm)
All dimension given in mm,
unless specified otherwise

PARTS LIST		
ITEM	QTY	PART NUMBER
1	1	1



Designed by mehta	Checked by	Approved by	Date		Date	
					11-11-2025	
			bigwall 1-1			
			Edition		Sheet 1 / 1	

2

1

Scale = 1:2
Dimension = 149x50x10
Material = M.S. Sheet(10mm)
Process = Conventional (Drilling, Cutting)
All dimension given in mm,
unless specified otherwise

PARTS LIST		
ITEM	QTY	PART NUMBER
1	1	2

50.00

13.00

149.00

10.00

DRAWN		TITLE		
Asus	11-11-2025			
CHECKED				
QA				
MFG				
APPROVED		SIZE	DWG NO	REV
		A	tallwall1	
		SCALE	1 / 2	SHEET 1 OF 1

B

B



A

A

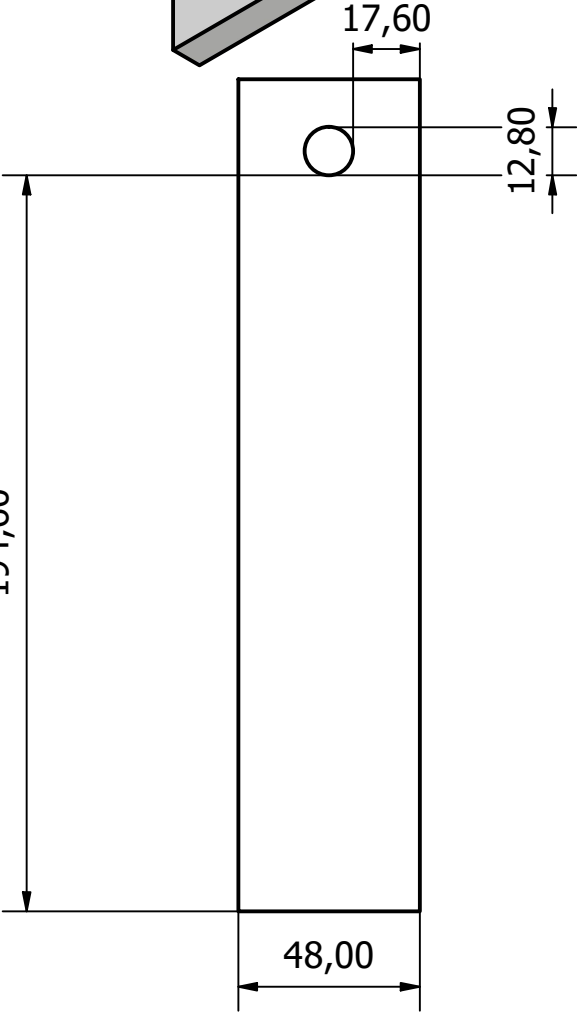
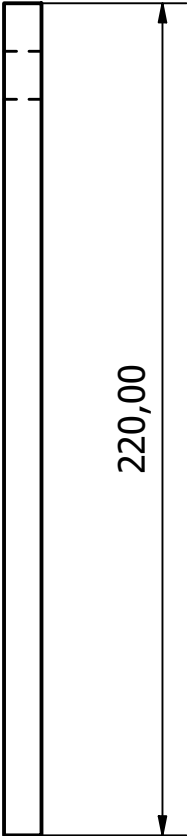
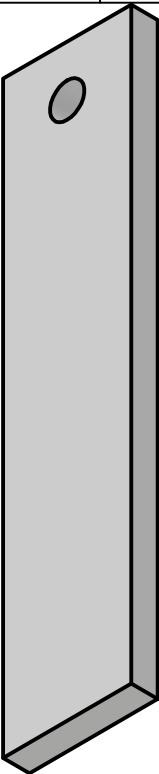
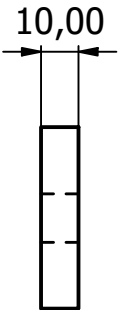


2

1

Scale=1:2
Dimension-220x48x10
Process-conventional(Drilling,
Miling,
Turning,Lathing)
All dimension given in mm unless specified otherwise
Material-M.S Sheet(10mm)

PARTS LIST		
ITEM	QTY	PART NUMBER
1	1	3



Designed by mehta	Checked by	Approved by	Date		Date	11-11-2025	
				tallwall2-2		Edition	Sheet 1 / 1

2

1

Scale = 1:2
Dimension = 160x49.5x10
Process =
Conventional(Drilling,Cutting)
Material = M.S. Sheet(10mm)
All dimension given in mm,
unless specified otherwise

PARTS LIST		
ITEM	QTY	PART NUMBER
1	1	4

49.50

17.00

160.00

10.00

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2

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Scale = 1:4

Process = Conventional(Lathing, Drilling,
Cutting)

Dimension = 420 x Ø14.35

Material = M.S. Rod

Threading = M16 x 2 - 6g

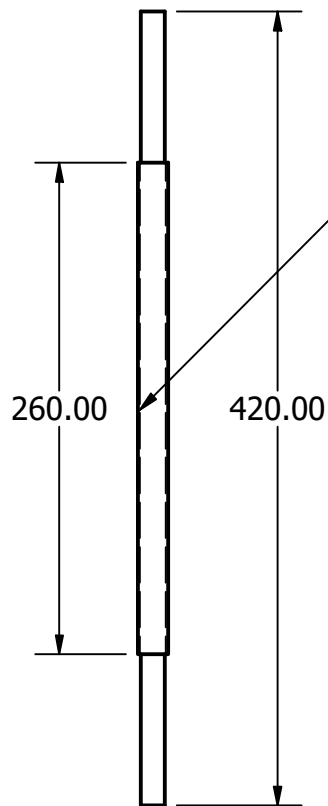
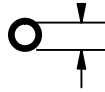
Thread Length = 260 (ISO Metric Thread)

All dimension given in mm,
unless specified otherwise

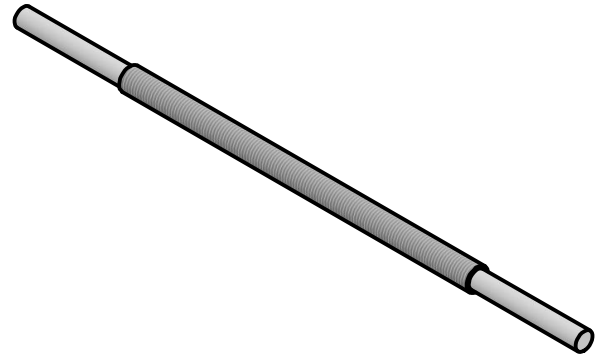
PARTS LIST

ITEM	QTY	PART NUMBER
1	1	5

14.35



M16x2 - 6g



B

B

A

A

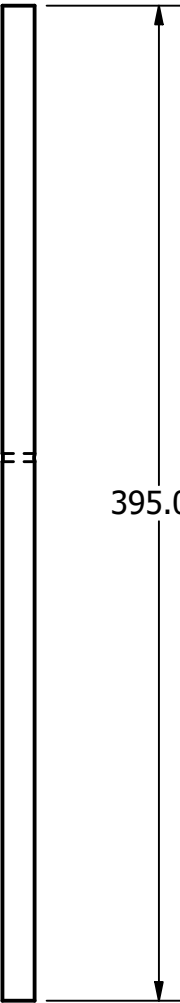
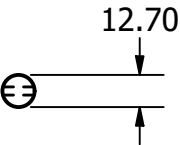
DRAWN	Asus	11-11-2025	TITLE			
CHECKED						
QA						
MFG						
APPROVED						
			SIZE		DWG NO	REV
			A		driveshaft1	
			SCALE	1 / 4	SHEET 1 OF 1	

2

1

PARTS LIST		
ITEM	QTY	PART NUMBER
1	1	7

Scale = 1:3
 Dimension = 395 x Ø12.70
 Material = M.S. Sheet
 Process = Turning/Lathing
 All dimension given in mm,
 unless specified otherwise



DRAWN	11-11-2025	TITLE		
Asus				
CHECKED				
QA				
MFG				
APPROVED		SIZE A DWG NO stationaryrod REV		
		SCALE	1 / 3	SHEET 1 OF 1

Scale-1:2

Dimension-145x70x52

Process-Conventional(Drilling,
milling,turning,lathing)

Material-Derlin

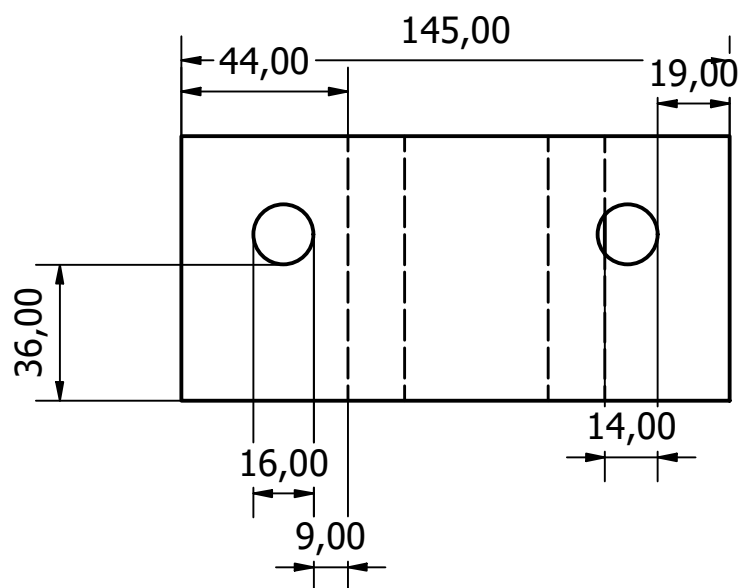
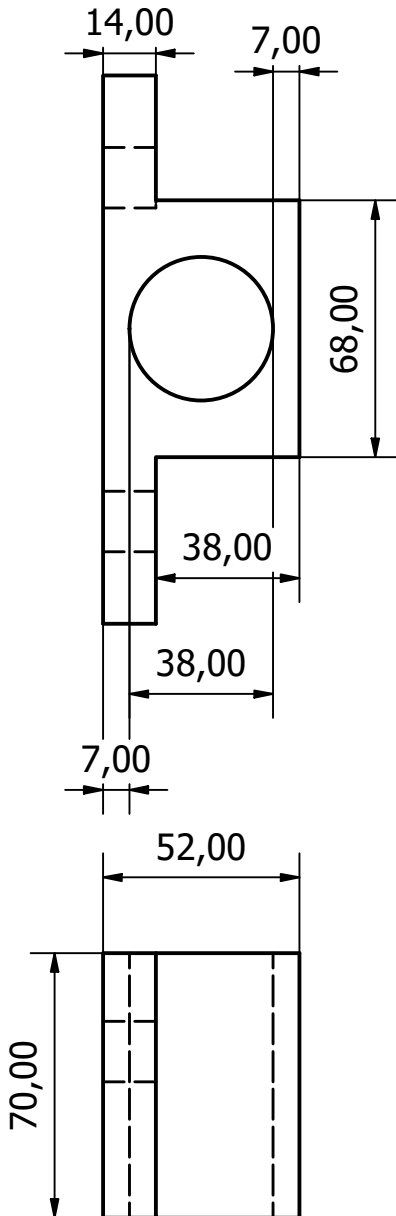
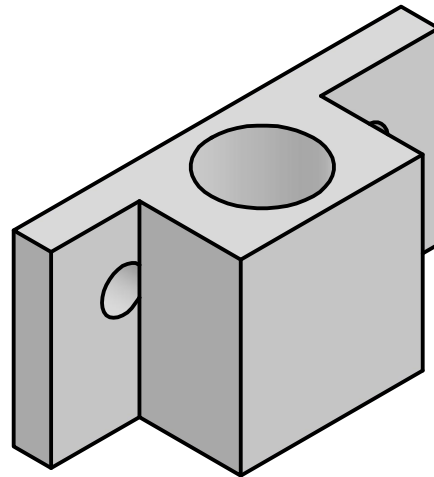
Threading: M16(Both 16 mm Holes)

All dimension are in mm unless specified otherwise

PARTS LIST

ITEM	QTY	PART NUMBER
1	1	8

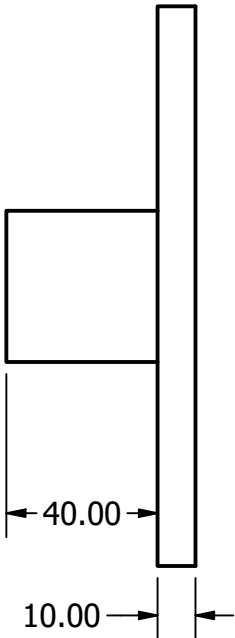
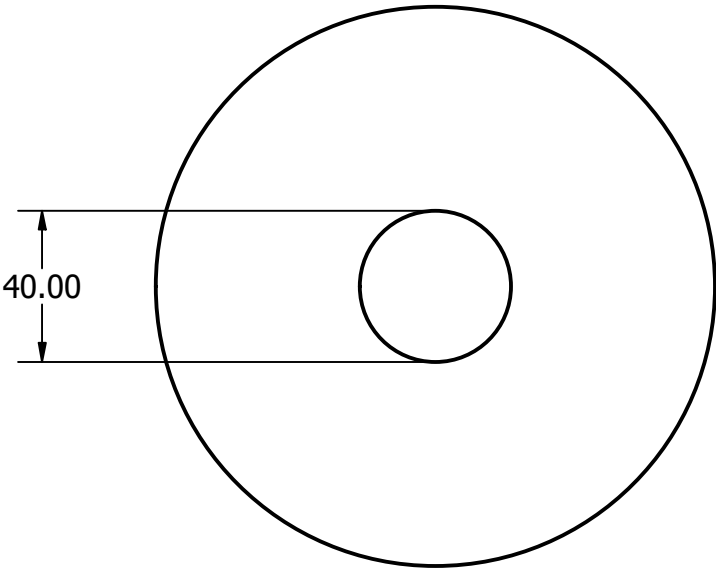
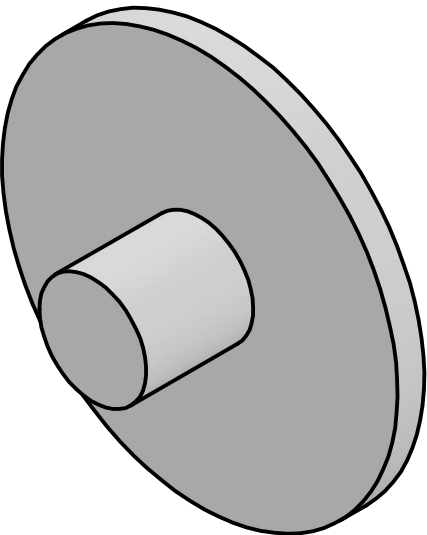
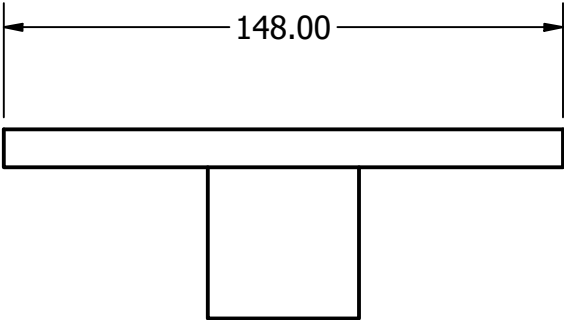
VIEW3 (1 : 2)



Designed by mehta	Checked by	Approved by	Date	Date 11-11-2025	
			carriage 8-8		
			Edition	Sheet 1 / 1	

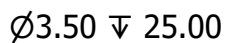
Scale = 1:2
Dimensions = 148x148x40
Material = PLA
Process = 3D Printing
All the dimension are in mm,
unless specified otherwise

PARTS LIST		
ITEM	QTY	PART NUMBER
1	1	9



DRAWN	Asus	11-11-2025	TITLE			
CHECKED						
QA						
MFG						
APPROVED			SIZE	DWG NO		REV
			A	Rough_Disc_Final		
			SCALE	1 / 2	SHEET 1 OF 1	

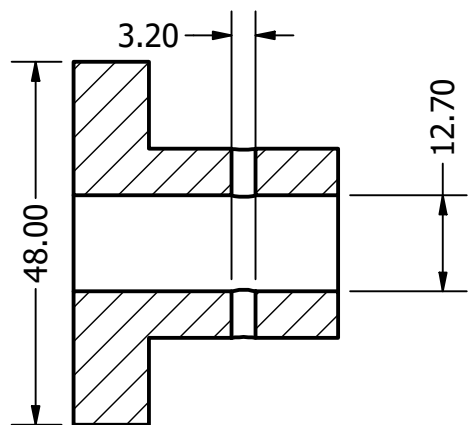
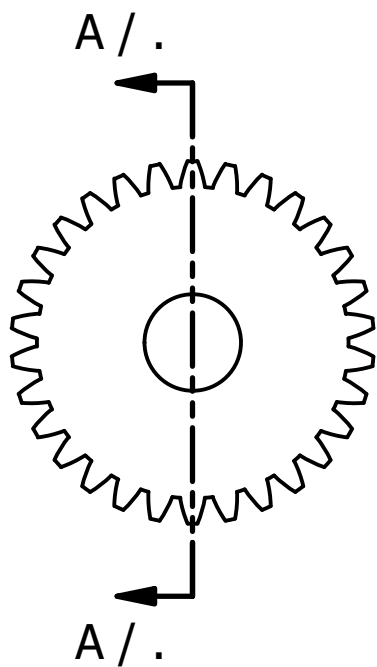
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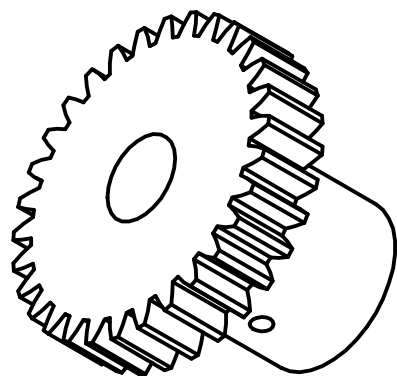
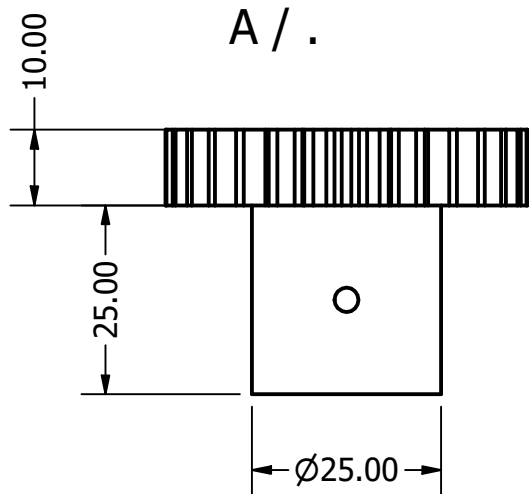
A

1

PART - 11



SECTION A-A
SCALE 1 : 1 / .



Scale 1 : 1

All dimensions in mm, unless
specified otherwise

Quantity: 2

Material: M.S. Rod

Process: Conventional (Drilling, Milling,
Turning/Lathing, Tooth Cutting)

No. of Teeth (n): 30

Module (m): 1.5

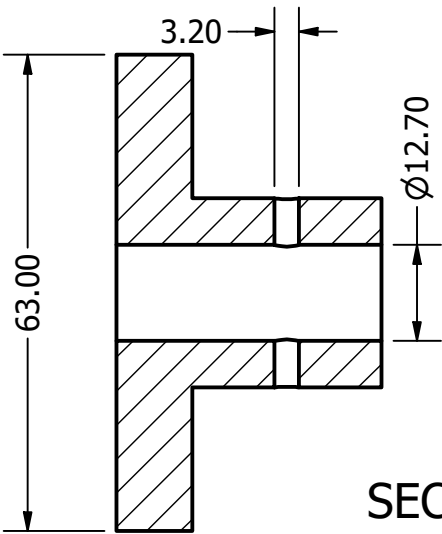
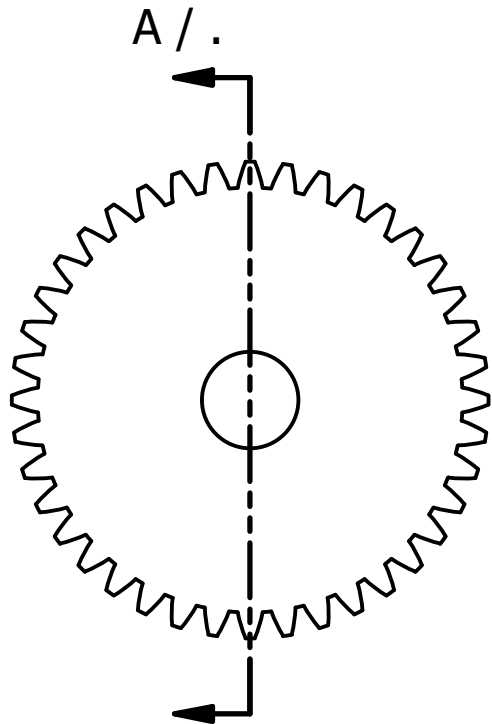
Outer Diameter (OD) = $m(n+2)$: 48

Rod Diameter (ID): 12.7

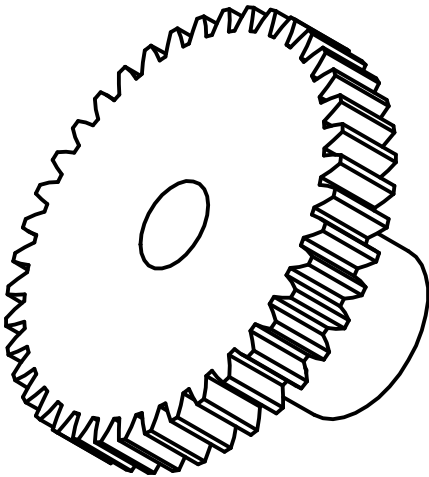
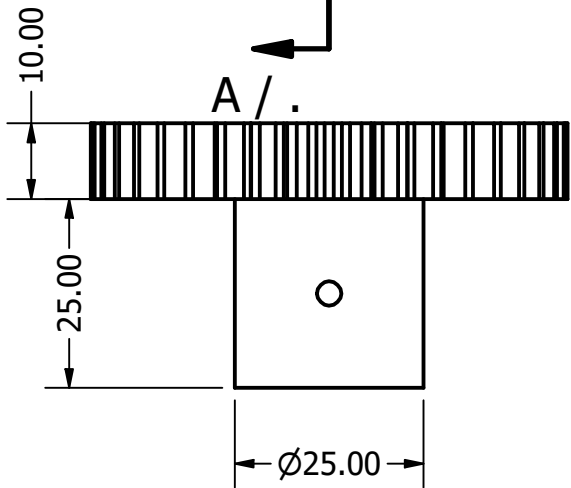
Indexing $(40/n)$: $1 \frac{1}{3}$

DRAWN Mukund Advaith	11-11-2025	TITLE			
CHECKED					
QA					
MFG					
APPROVED		SIZE			
		A4		DWG NO	
				spur1	
		SCALE		REV	
		1 : 1		SHEET 1 OF 1	

PART - 12



SECTION A-A
SCALE 1 : 1 / .



Scale 1 : 1
All dimensions in mm, unless
specified otherwise
Quantity: 1
Material: M.S. Rod
Process: Conventional (Drilling, Milling,
Turning/Lathing, Tooth Cutting)

No. of Teeth (n): 40
Module (m): 1.5
Outer Diameter (OD) = $m(n+2)$: 63
Rod Diameter (ID): 12.7
Indexing (40/n): 1

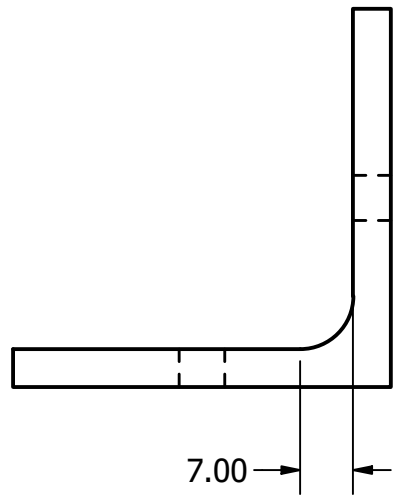
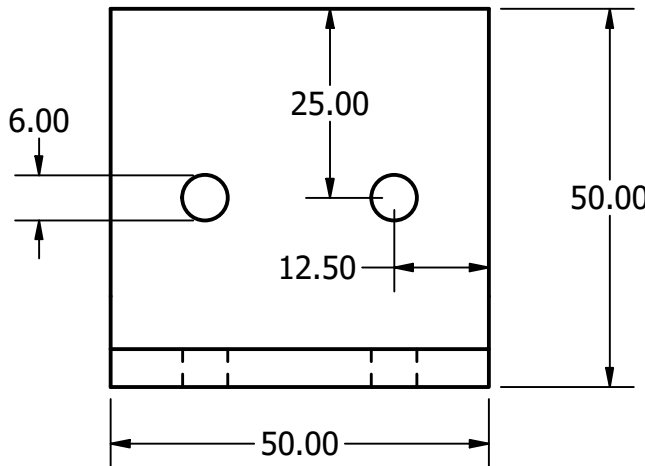
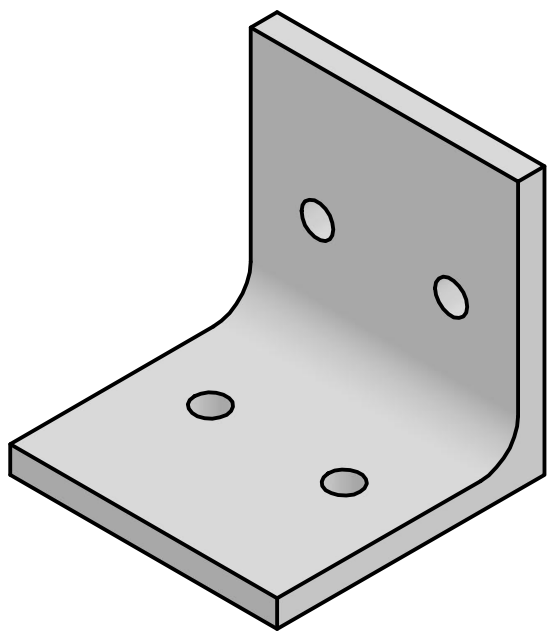
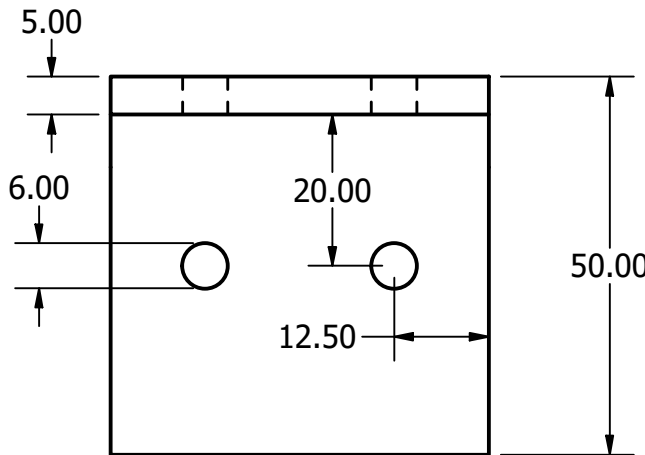
DRAWN	Mukund Advaith	11-11-2025	TITLE			
CHECKED						
QA						
MFG						
APPROVED			SIZE			
			A4		DWG NO	
					spur2	
			SCALE		REV	
			1 : 1		SHEET 1 OF 1	

All dimensions are in mm unless specified otherwise

Designed by mehta	Checked by	Approved by	Date	Date	
		motorholder 13 13		Edition	Sheet 1 / 1

Scale = 1:1
Process = Conventional (Drilling,
Milling, Cutting, Turning, Lathing)
Dimension = 50x50x5
Material = M.S. Angle
All dimensions given in mm,
unless specified otherwise

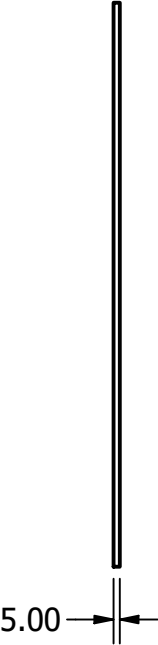
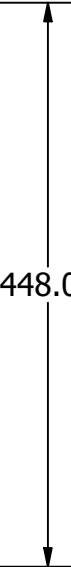
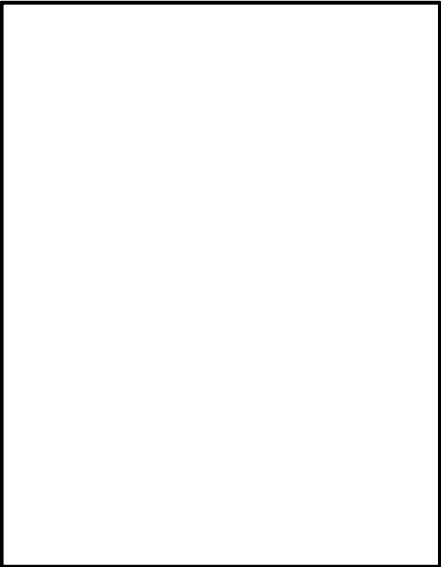
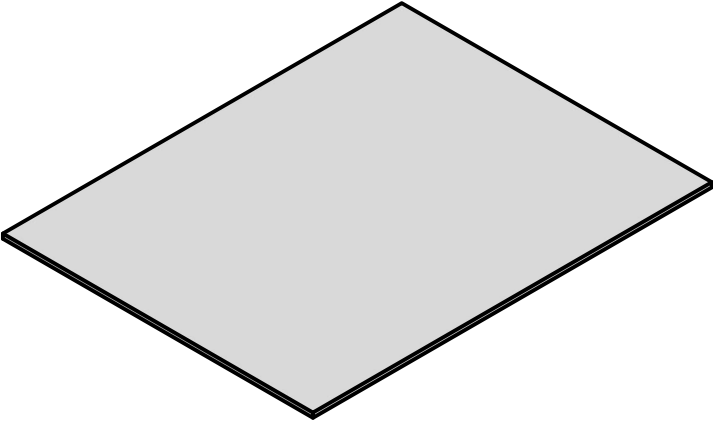
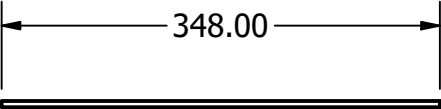
PARTS LIST		
ITEM	QTY	PART NUMBER
1	1	7



DRAWN	Asus	11-11-2025	TITLE		
CHECKED					
QA					
MFG					
APPROVED					
			SIZE	DWG NO	REV
			A	lclamp	
			SCALE	1 : 1	SHEET 1 OF 1

Scale = 1:6
Process = Drilling, Cutting
Material = M.S. Sheet
Dimension = 448x348x5
All dimensions given in mm,
unless specified otherwise

PARTS LIST		
ITEM	QTY	PART NUMBER
1	1	16



DRAWN	Asus	11-11-2025	TITLE			
CHECKED						
QA						
MFG						
APPROVED						
			SIZE		DWG NO	REV
			A		base	
			SCALE	1 / 6		SHEET 1 OF 1

Project objective:

The objective of this project is to design and develop MECHANICAL INTEGRATOR using gears, driving shaft, motor and other mechanical components . All these components are made by using methods which have learnt in the course of TA212. The system converts an integral. It demonstrates the principles of mechanical computation ,motion transfer and other analog data processing.

Motivation

The primary motivation for this project was to apply our engineering skills to a significant, real-world problem. We sought to move beyond a purely academic exercise and create a functional machine with practical applications in fields like agriculture or the military.

Our research identified a critical inefficiency in modern agriculture: **traditional furrow irrigation**. This common method, while inexpensive, is extremely wasteful. It is estimated that 30-50% of water is lost to uncontrolled runoff and uneven deep percolation. This leads to inconsistent soil moisture, which stresses plants, damages crop health, and ultimately reduces the overall yield of a field.

The core of this waste is a problem of measurement. Irrigation is often controlled by **time**, but due to variable water pressure, the **flow rate** is never constant. The correct metric for a plant's needs is **volume**. To deliver a precise volume, one must continuously calculate:

$$\text{Total Volume} = \int \text{Flow Rate} \times d(\text{Time})$$

This is a calculus problem. This realization led us to our solution: the **Mechanical Integrator**. We were motivated by the challenge of building a robust, analog computer that could solve this exact equation mechanically, creating a "Mechanical Water-Bailiff" to precisely control water delivery.

This goal presented a significant technical challenge. We quickly discovered that a functional integrator is a "closed-form" system where every component is dimensionally interdependent. The design and its exact measurements were not known in advance and had to be discovered through a rigorous, iterative process of **trial and error**.

The required kinematic precision—the exact fit of the Carriage/Mover on its rod, the alignment of the Drive Shaft through the Support Stands, and the consistent contact of the Stationary Wheel on the Base Disk—became the central focus. Our motivation was therefore twofold: to first solve the practical problem of water waste by designing a device, and to then overcome the complex manufacturing and assembly challenges required to *build* that precise analog computer, applying the full scope of our TA212-course skills.

Acknowledgement:

We would like to express our heartfelt gratitude to the course instructor SARVESH MSHRA and the **Department of Mechanical Engineering** for giving us the opportunity to do this project. We would also like to thank our **project guide, Mr Arun Kumar Dubey** for his constant guidance, encouragement, and support throughout the completion of our project on *Mechanical Integrator*. We also sincerely thank the **laboratory staff and technical team** for their assistance in fabricating and testing the model.

This project helped us gain practical knowledge of **mechanical computation, gear mechanisms, and motion transmission systems**, and enhanced our understanding of **integration principles through mechanical means**.

MOTOR TORQUE CALCULATIONS

Motor connected to gears:

The motor is connected to spur gear (n=40), which turns two separate spur gears (n=30), each connected to its own drive shaft. The total torques of motor is split across both the drive shafts.

$$I_{total\ driven} = I_{spur\ gear\ (40)} + 2I_{spur\ gear\ (30)} \\ + I_{drive\ shaft\ 1} + I_{drive\ shaft\ 2}$$

(I is Moment of Inertia)

$$I_{spur\ gear} = \frac{1}{2}\rho\pi(R_1^4 \cdot L_1 + R_2^4 \cdot L_2 - R_h^4 \cdot L_h)$$

$$I_{drive\ shaft} = \frac{1}{2}\rho\pi R^4 L$$

Where ρ = Density of Mild Steel

R_1 = Outer radius of gear teeth part

R_h = Radius of gear hole

L_1 = Length of gear teeth part

L_h = Length of Gear hole

R_2 = Outer radius of gear collar

R = Outer radius of driveshaft

L = Length of driveshaft

L_2 = Length of gear collar

After putting in values from schematics, we get

$$I_{spur\ 40} = 0.128 \times 10^{-3} \text{ kgm}^2, \quad I_{spur\ 30} = 0.477 \times 10^{-3} \text{ kgm}^2, \\ I_{driveshaft\ 1} = 1.868 \times 10^{-3} \text{ kgm}^2, \quad I_{driveshaft\ 2} = 2.120 \times 10^{-3} \text{ kgm}^2,$$

Combining as per above formula, we get,

$$I_{total\ driven} = 2.633 \times 10^{-3} \text{ kgm}^2,$$

We know that,

$$\tau (\text{Torque}) = I_{total\ driven} \times \alpha (\text{Angular Acceleration})$$

By practical observation, we found that Rectangular High Torque Motor (3 N-m) provided enough torque to move the system of given moment of inertia with a satisfactory α .

Motor connected to Big Disk

As big disk is a 3D printed component, the load on the motor is less and any suitable motor can be used. We are using normal DC motor (2 N-m).

Cost Estimation for Mechanical Integrator Prototype

Component / Service	Details	Cost (₹)
Mild Steel Parts	14.9 kg @ ₹90/kg	1,341.00
Nuts & Bolts (25 pairs)	1.25 kg @ ₹130/kg	162.50
Derlin Part	0.157 kg @ ₹300/kg	47.00
Drilling (Machine)	6 hours @ ₹100/hr	600.00
Lathe (Machine)	12 hours @ ₹100/hr	1,200.00
Milling (Machine)	3 hours @ ₹100/hr	300.00
Cutting (Machine)	1 hour @ ₹100/hr	100.00
3D Printing (Machine)	2.5 hours @ ₹300/hr	750.00
Motors		₹1,300 + ₹600 = 1,900.00

| Total Estimated Cost = ₹6,400.50 ≈ ₹6,400 |

Notes

- The integrator consists of two 3D-printed parts: one large disk and one small disk. PLA material cost is excluded from the estimation (only printing time cost included).
- A high-torque rectangular motor (₹1,300) and a single DC motor (₹600) are used.
- 25 nut-bolt pairs were used (average pair ≈ 0.05 kg → 1.25 kg total).
- Derlin part weighs approximately 0.157 kg.
- Mild steel part weight was estimated by subtracting the weight of the other components from a total system weight of ~17 kg.
- Total project weight: approximately 17 kg.
- Labor cost is not included in this calculation.